

# The Wang Method Series

## Notebook #4 (Version 1.0)

### Wang's Simple L3E Pairing Method

*A Look Ahead Method for Solving the Last Three Edge Pairs of the 4×4 Rubik's Cube*

Laung-Terng (LT) Wang

August 1, 2026

#### 1. Introduction

After completing the first five edge pairs using the **3-2-3 edge pairing strategy**, the last three edge pairs (**L3E**) remain unsolved. Efficiently solving these last three edge pairs is one of the key ideas of the **Yau Method**.

This notebook presents **Wang's Simple L3E Pairing Method**, a beginner-friendly approach based on **Look Ahead**. Instead of immediately pairing the most obvious edge, which may unintentionally reduce the problem to a last two edge pairs (**L2E**) configuration, the solver first examines the complete L3E configuration and predicts the result of each possible pairing.

Two complementary search paths are used to determine whether an edge can be paired directly or should first be transformed by a **Flip** or **Swap** operation. The selected operation preserves the **L3E** structure so that the solver can continue solving all three remaining edge pairs systematically.

Figures 4.1 and 4.2 summarize the complete Look Ahead decision process.

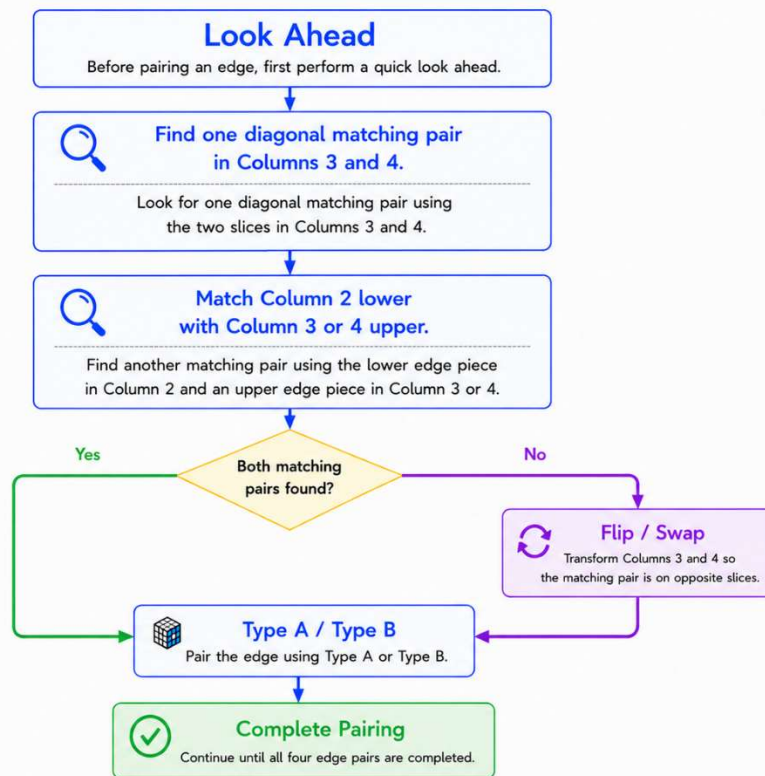


Figure 4.1. Look Ahead flowchart: Search Path 1.

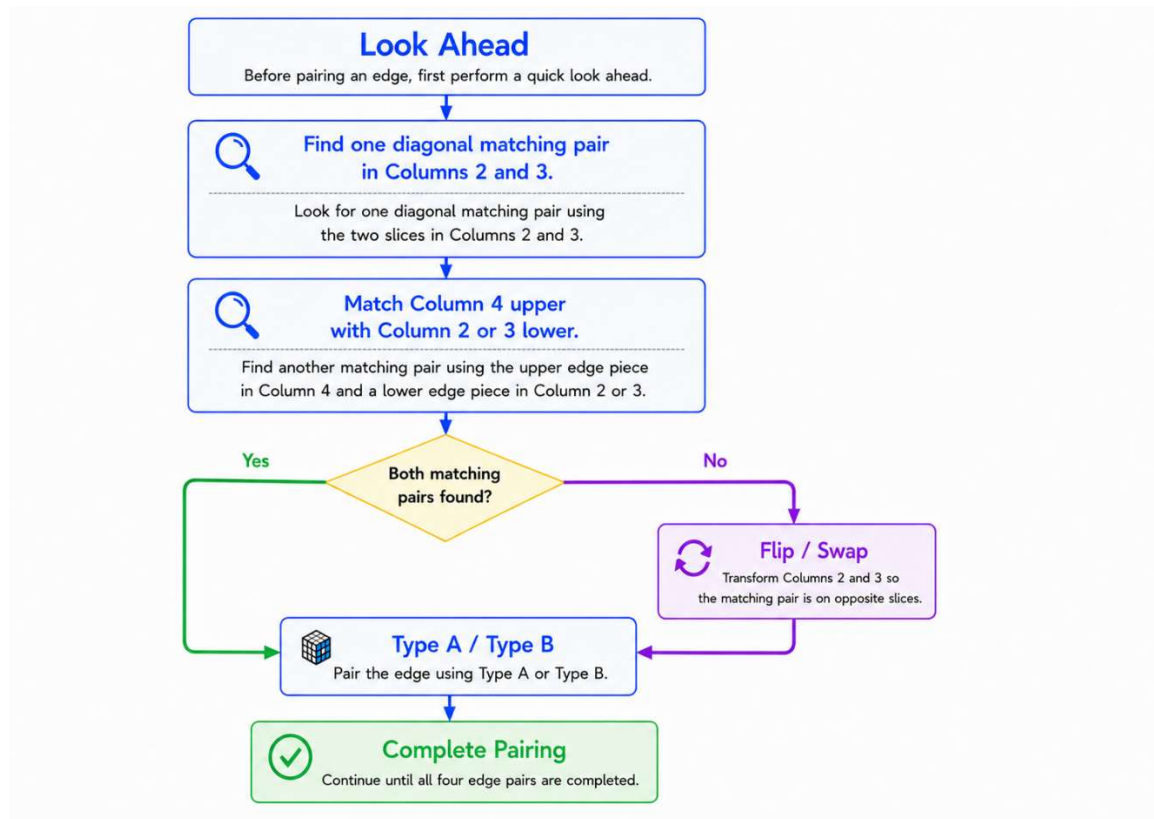


Figure 4.2. Look Ahead flowchart: Search Path 2.

## 2. Look Ahead

Before pairing an edge, first perform a quick **Look Ahead**.

Begin with **Search Path 1** (Figure 4.1).

- Find one diagonal matching pair in **Columns 3 and 4**.
- Then match the lower edge piece in **Column 2** with an upper edge piece in **Column 3 or 4**.

If both matching pairs are found, pair the edge directly using **Type A** or **Type B**.

If Search Path 1 is unsuccessful, try the complementary **Search Path 2** (Figure 4.2).

- Find one diagonal matching pair in **Columns 2 and 3**.
- Then match the upper edge piece in **Column 4** with a lower edge piece in **Column 2 or 3**.

Again, if both matching pairs are found, pair the edge directly using **Type A** or **Type B**.

Otherwise, perform a **Flip** or **Swap**, then complete the pairing using **Type A** or **Type B**.

This Look Ahead strategy considers the complete **L3E** configuration before any pairing operation is performed. Rather than simply solving the immediately visible edge pair, it selects a procedure that preserves the **L3E** structure and avoids unnecessarily inducing an **L2E** configuration.

### 3. Type A Pairing

Type A uses an upper-row shift.

Type A is one of the two fundamental pairing procedures of Wang's Simple L3E Pairing Method. The detailed five-step procedure is illustrated in Figure 4.3. Once Type A is mastered, the remaining configurations can be solved by first transforming them into Type A and then applying this procedure.

The complete Type A pairing procedure requires **five moves**.

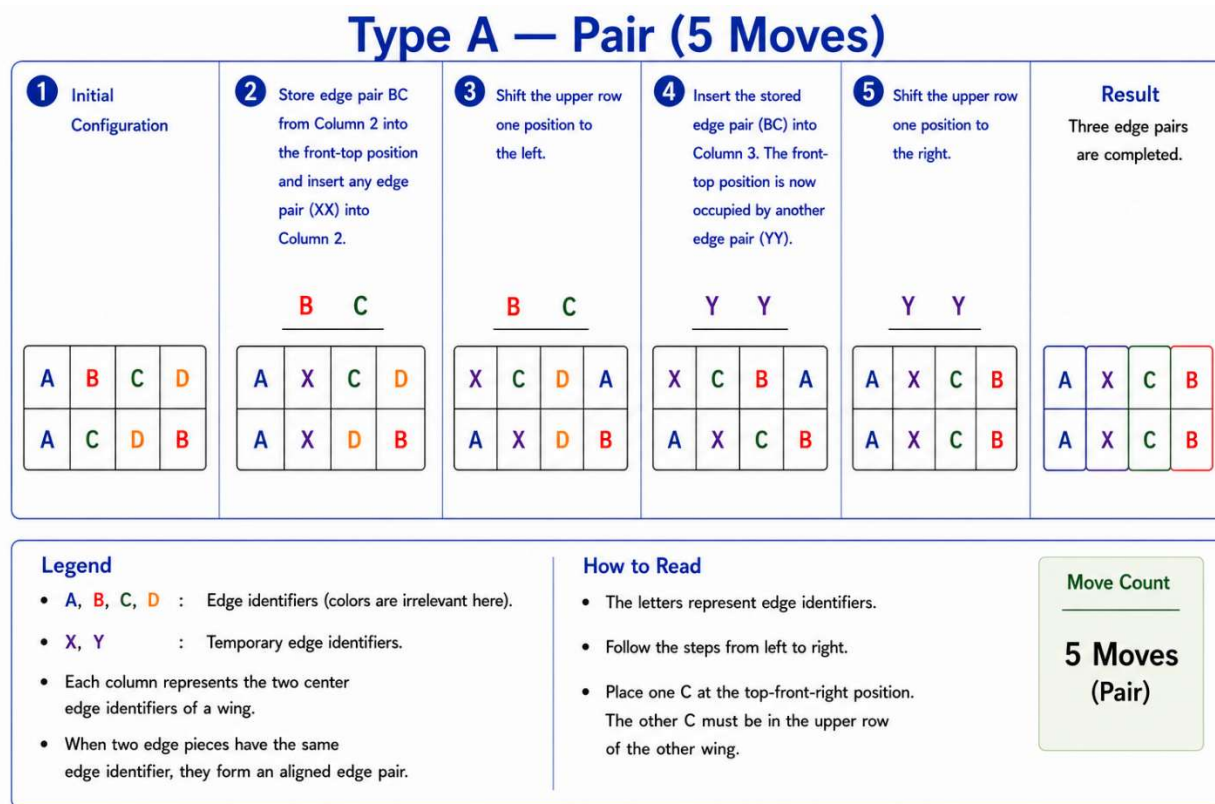


Figure 4.3: Type A — Pair (5 Moves).

## 4. Type B Pairing

Type B uses the same store-and-insert concept as Type A, but the lower row is shifted instead of the upper row.

Type B is the second fundamental pairing procedure of Wang's Simple L3E Pairing Method. Like Type A, it assumes that the two matching **C edge pieces** have already been positioned on opposite slices.

The complete Type B pairing procedure also requires **five moves**.

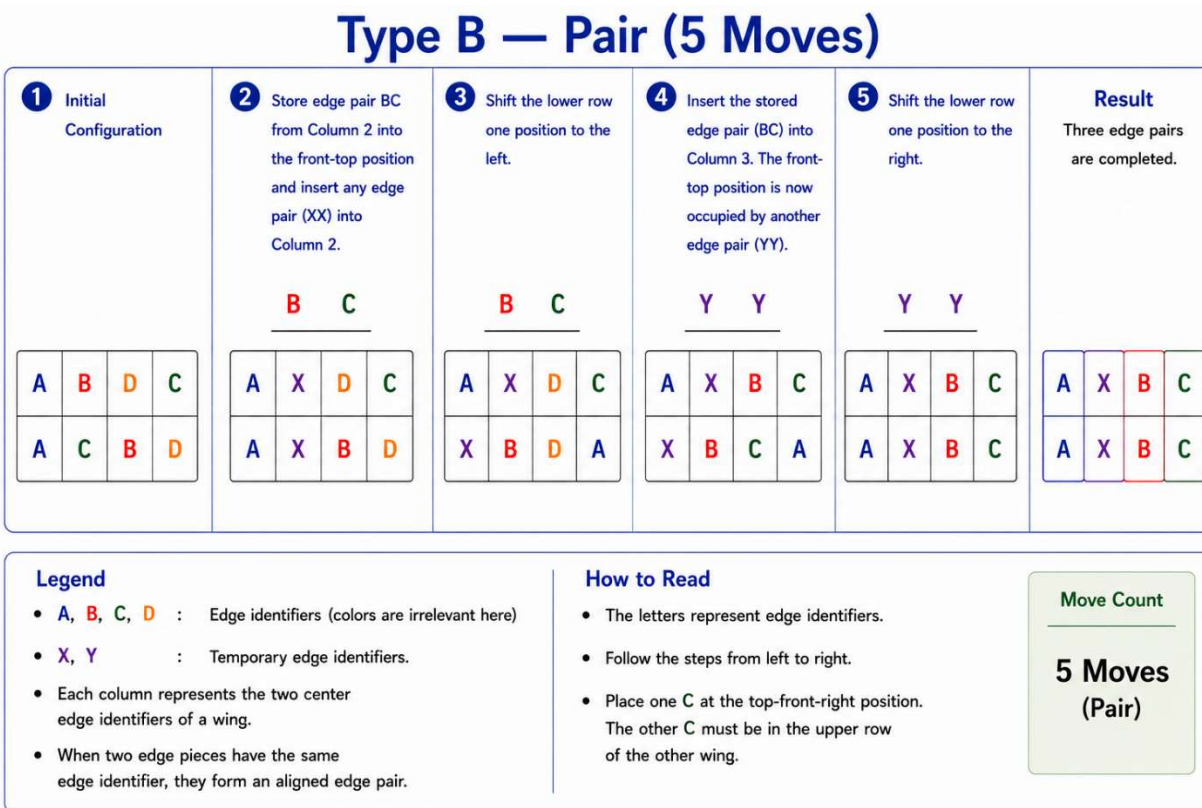


Figure 4.4: Type B — Pair (5 Moves).

## 5. Flip + Pair Transformations

Flip + Pair is the primary transformation used in Wang's Simple L3E Pairing Method.

A flip separates the two matching **C edge pieces** that initially occupy the same slice. The resulting configuration becomes either **Type A** or **Type B**, and can then be solved using the corresponding five-move pairing procedure.

Although a wing can be flipped in fewer moves, Wang's Simple L3E Pairing Method intentionally uses the **seven-move Flip** because it preserves the remaining prepared edge positions. After the last three edge pairs are completed, the solver can proceed directly to the 3×3 stage and begin **CFOP** with the Cross. The shortest move sequence is not always the best solution.

The complete **Flip + Pair** sequence therefore requires **twelve moves**.

## 5.1 Type A — Flip + Pair

For the Type A transformation, one wing is flipped so that the two matching **C** edge pieces move onto opposite slices. The resulting configuration is identical to the initial configuration of **Type A — Pair**.

The Type A pairing procedure is then applied.

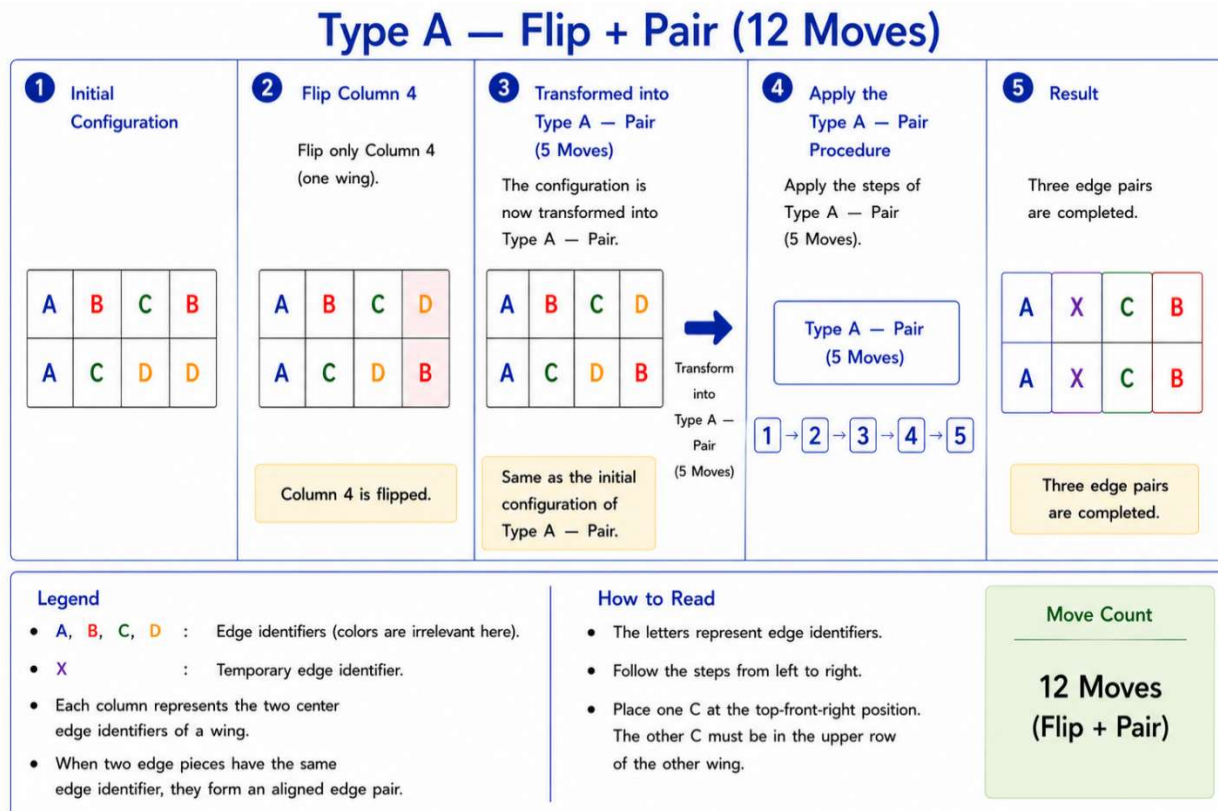


Figure 4.5: Type A — Flip + Pair (12 Moves).

## 5.2 Type B — Flip + Pair

For the Type B transformation, one wing is flipped to separate the two matching **C** edge pieces. The resulting configuration becomes **Type B — Pair**.

The Type B pairing procedure is then applied.

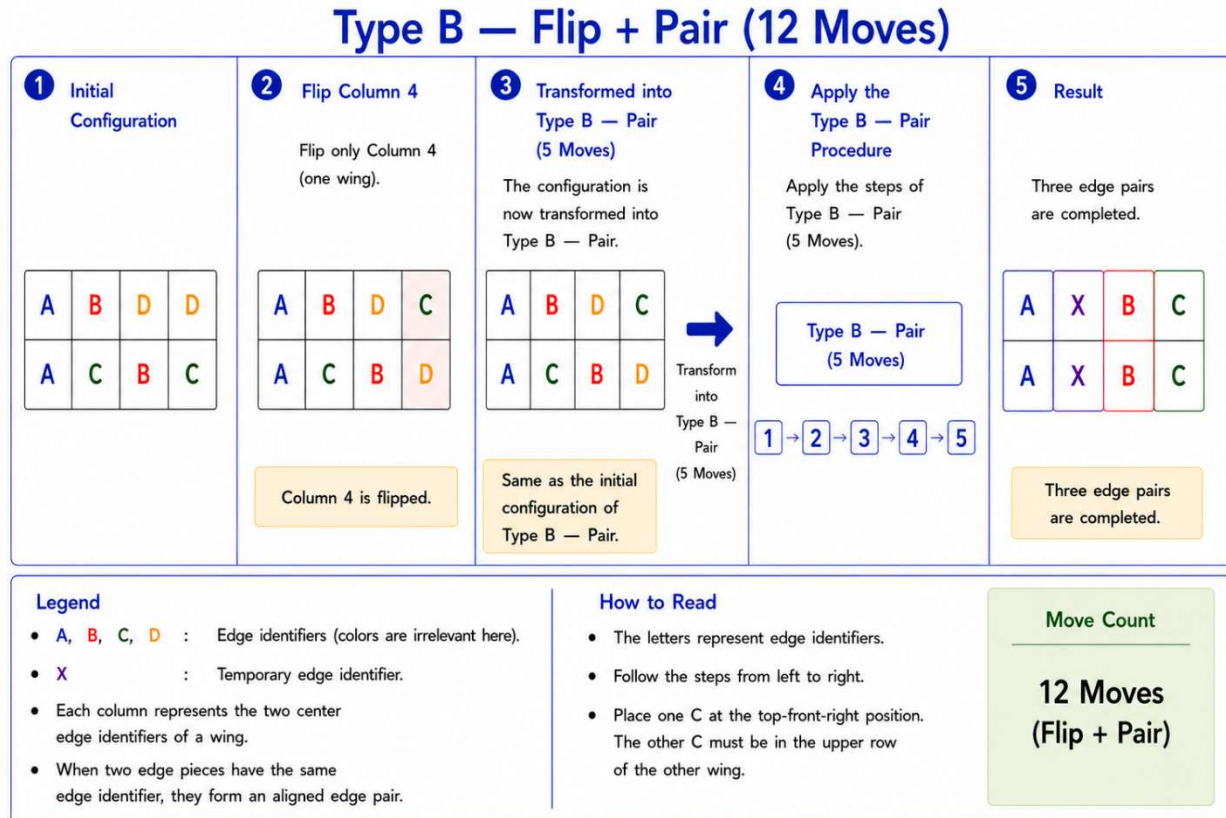


Figure 4.6: Type B — Flip + Pair (12 Moves).

## 6. Type C and Type D Recognition

Type C and Type D represent the two principal same-slice arrangements.

In both types, the two matching **C edge pieces** initially occupy the same slice. Therefore, the solver should not directly complete another edge pair without first transforming the configuration.

### 6.1 Type C — Flip + Pair

In Type C, flipping the indicated wing transforms the configuration into **Type A — Pair**. The Type A pairing procedure then completes all three edge pairs.

Thus, Type C is solved by:

**Flip → Type A — Pair**

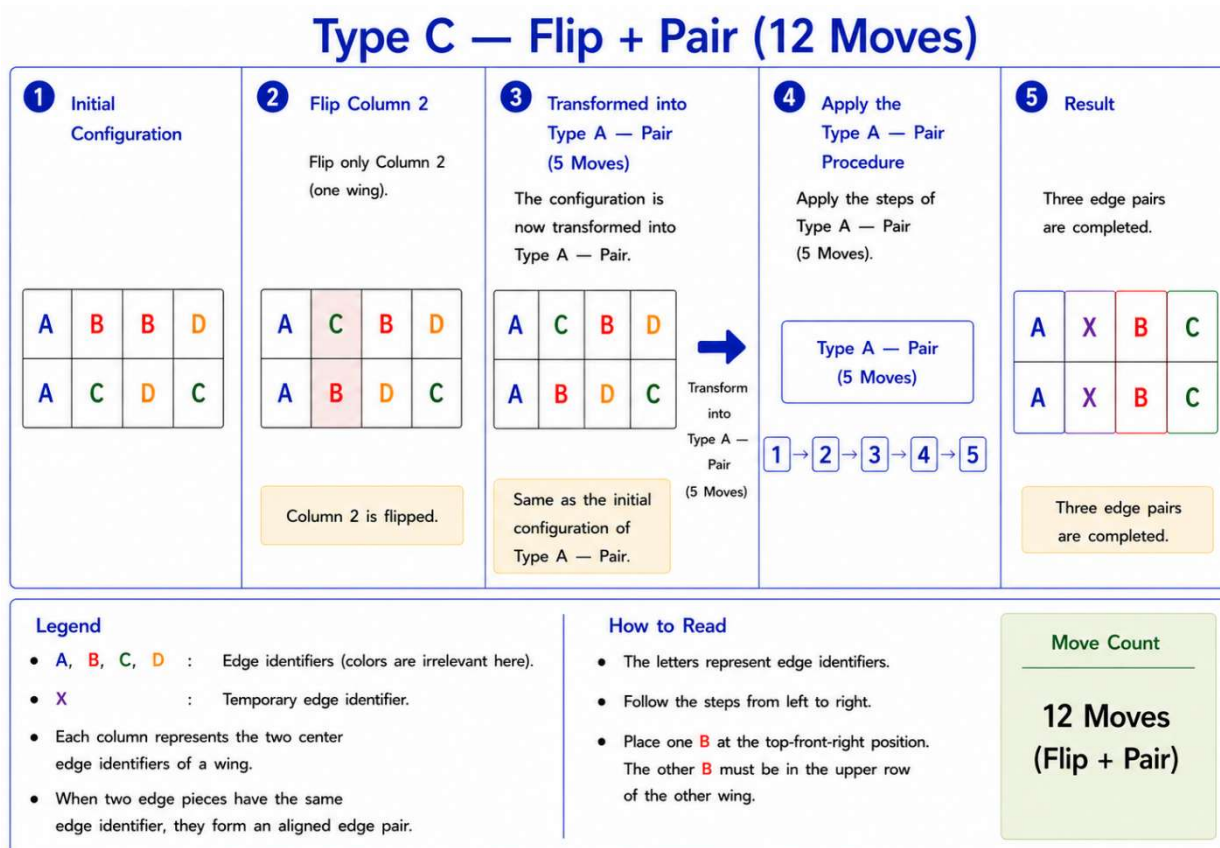


Figure 4.7: Type C — Flip + Pair (12 Moves).

## 6.2 Type D — Flip + Pair

In Type D, flipping the indicated wing transforms the configuration into **Type B — Pair**. The Type B pairing procedure then completes all three edge pairs.

Thus, Type D is solved by:

**Flip → Type B — Pair**

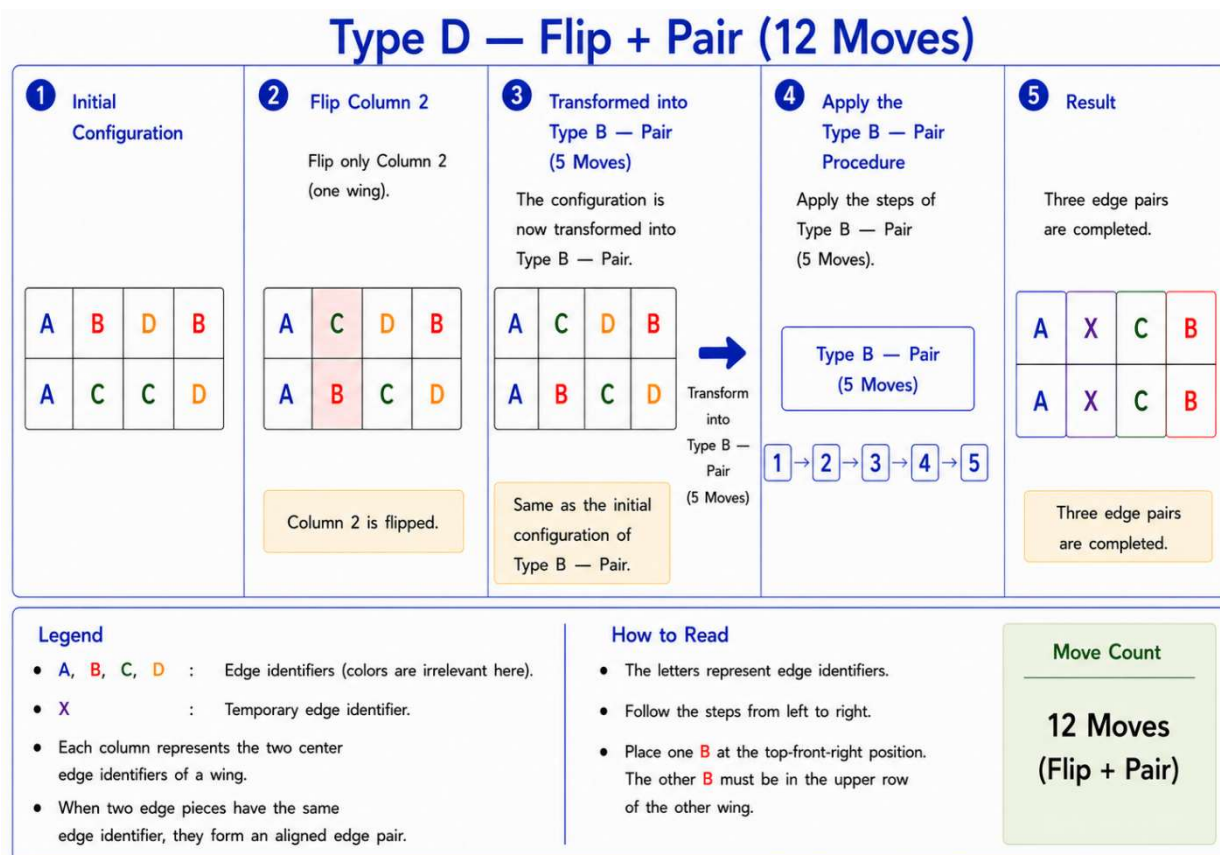


Figure 4.8: Type D — Flip + Pair (12 Moves).

No additional pairing algorithm is required for Type C or Type D.

## 7. Optional Swap + Pair Procedures

Flip + Pair requires only twelve moves and is therefore the preferred procedure for beginners.

However, the shortest move count does not always produce the shortest execution time for an experienced speed cuber.

When the edge pair to be preserved is already conveniently positioned at the front-top, a speed cuber may prefer to perform a Swap instead of a Flip. Although Swap + Pair requires fourteen moves, it may provide a smoother execution with fewer pauses for recognition or regripping.

Like Flip, a Swap can also be performed in fewer moves. However, Wang's Simple L3E Pairing Method intentionally uses the **nine-move Swap** because it preserves the remaining prepared edge positions. After the last three edge pairs are completed, the solver can proceed directly to the 3×3 stage and begin the standard 3×3 **CFOP** method with the Cross.

For this reason, Wang's Simple L3E Pairing Method includes **Swap + Pair** as an optional speed-cubing alternative.

The two approaches serve different purposes:

- **Flip + Pair:** fewer moves and easier for beginners to learn.
- **Swap + Pair:** an optional alternative that may be faster in actual execution when the desired edge pair is already positioned at the front-top.

The choice should therefore be based not only on move count, but also on the current cube position and the solver's execution preference.

### 7.1 Type A — Swap + Pair

In Type A, swapping the indicated edge pieces transforms the configuration into **Type A — Pair**. The corresponding five-move pairing procedure then completes all three edge pairs.

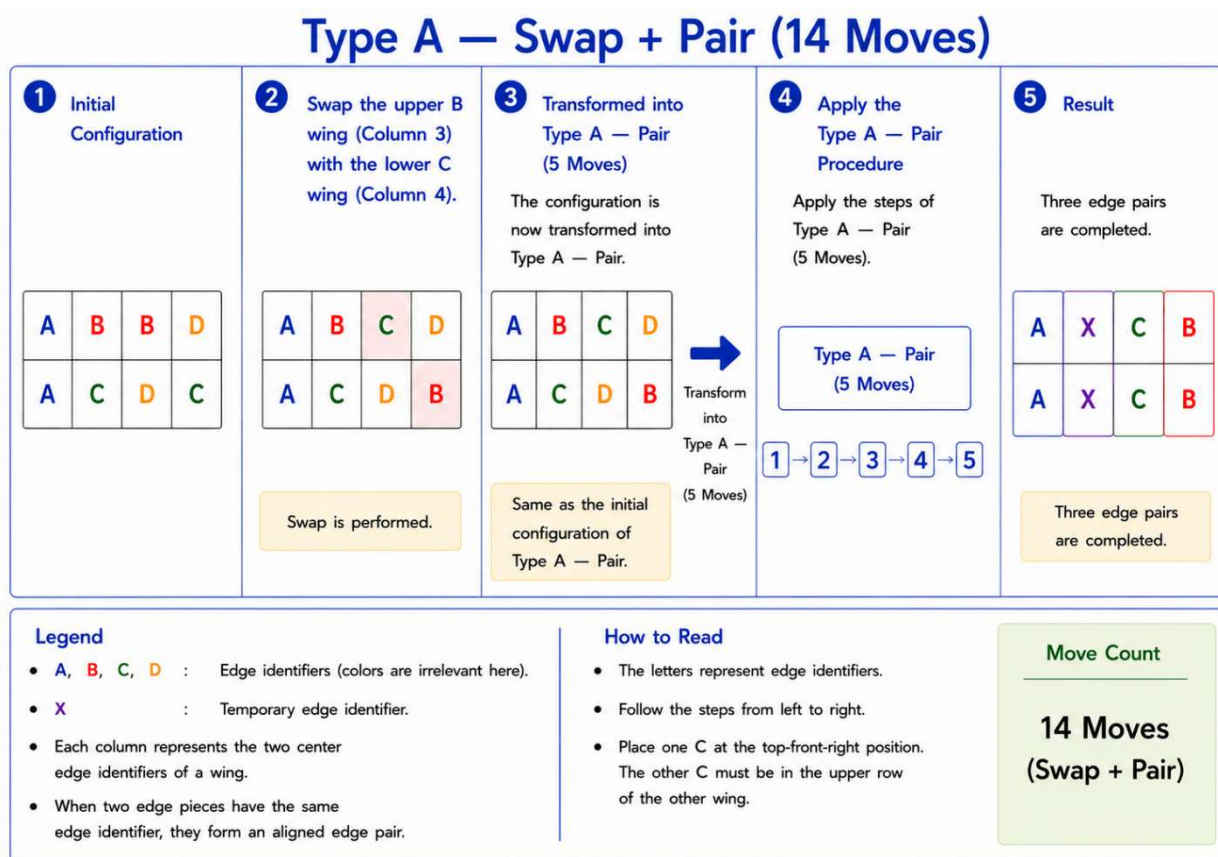


Figure 4.9: Type A — Swap + Pair (14 Moves).

## 7.2 Type B — Swap + Pair

In Type B, swapping the indicated edge pieces transforms the configuration into **Type B — Pair**. The corresponding five-move pairing procedure then completes all three edge pairs.

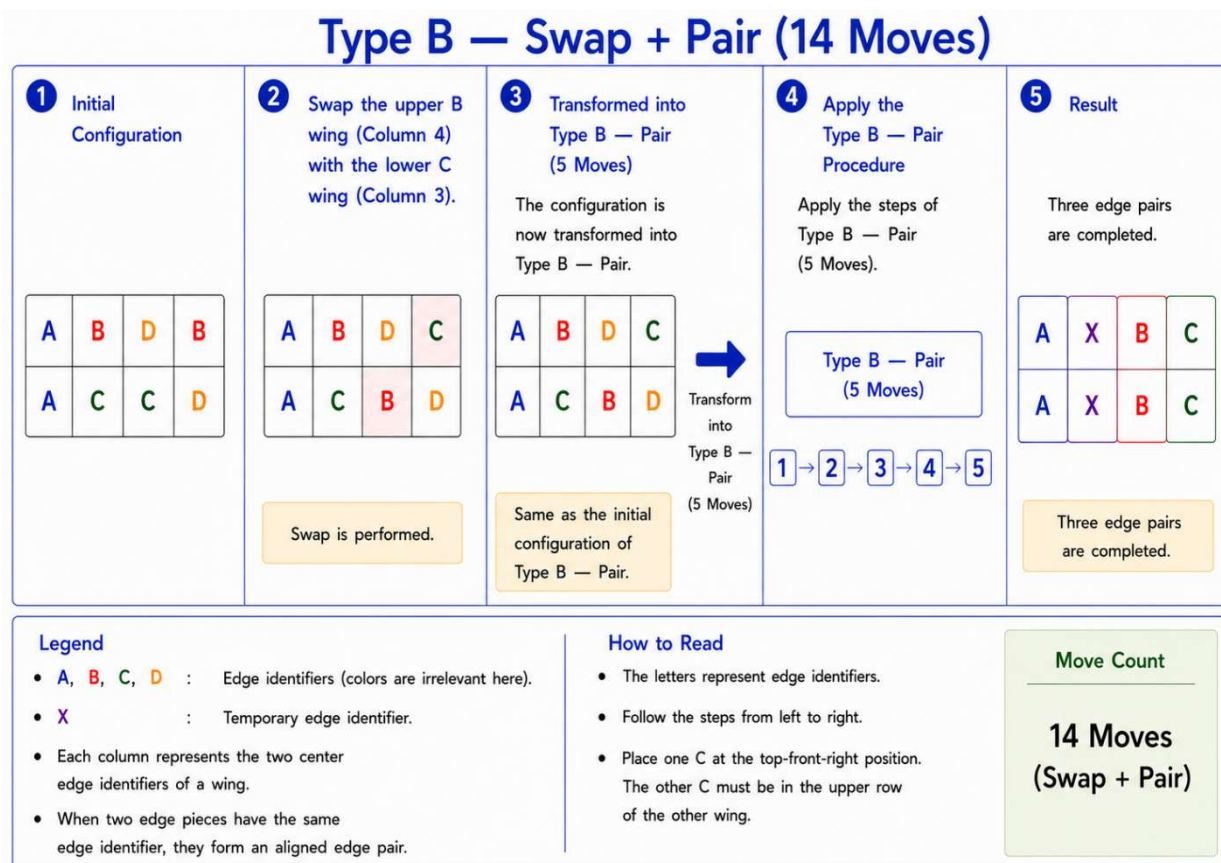


Figure 4.10: Type B — Swap + Pair (14 Moves).

## 8. Summary

The complete pairing strategy can be summarized as follows.

- Perform **Look Ahead**.
- Try **Search Path 1**.
- If both matching pairs are found, apply **Type A** or **Type B**.
- If unsuccessful, try the complementary **Search Path 2**.
- If both matching pairs are found, apply **Type A** or **Type B**.
- Otherwise, perform a **Flip** or **Swap**, then apply **Type A** or **Type B**.
- Repeat the process until all three remaining edge pairs are completed.

By looking ahead before every pairing, this method preserves the **L3E** structure and avoids unnecessarily inducing an **L2E** configuration.

Only **three move counts** need to be remembered throughout Wang's Simple L3E Pairing Method:

- **5 moves** for direct pairing (**Type A** or **Type B**),
- **12 moves** for **Flip + Pair**, and
- **14 moves** for **Swap + Pair**.

Unlike the traditional Yau L3E pairing approach, this method does **not** require memorizing numerous algorithms or exceptional cases. Instead, it combines two complementary search paths with four simple procedures—**Type A**, **Type B**, **Flip**, and **Swap**—to provide a simple, systematic, and easy-to-learn solution while preserving the principal advantage of the Yau edge pairing method: the solved white cross is maintained, allowing the solver to proceed directly to the 3×3 **CFOP** stage without rebuilding the white cross.

---

## Acknowledgments

The author gratefully acknowledges the Rubik's Cube community for generously sharing educational videos, tutorials, and solving techniques online. Watching many of these videos provided valuable opportunities to observe different solving styles and inspired further exploration of simpler and more systematic approaches to the L3E pairing problem.

Wang's Simple L3E Pairing Method presented in this notebook was independently developed through the author's own analysis, classification, and optimization of L3E configurations.

---

## References

- [1] Feliks Zemdeg, *Yau Method for Solving the 4x4 Cube*, CubeSkills, 2020.
- [2] J Perm, *Learn How to Solve a 4x4 in 10 Minutes (Full Yau Method Tutorial)*, YouTube, January 29, 2018.
- [3] J Perm, *4x4 Yau Method: 3-2-3 Edge Pairing (+ Special Cases)*, YouTube, 2017.
- [4] Noah Richardson, *4×4 Last Edges Pairing (Beginner Friendly) | Final Edge Fix Explained*, YouTube, January 2026.